

and combining (3) and (4) establishes our claim that $c(\tau) = c(4\tau)$. Successive applications of this identity give $c(\tau) = c(4^k\tau)$, and since $q^{4^k} = e^{i\pi 4^k\tau} \rightarrow 0$ as $k \rightarrow \infty$, we conclude from (2) that $c(\tau) = 1$. This proves the theorem.

The product formula for the function Θ specializes to its variant $\theta(\tau) = \Theta(0|\tau)$, and this provides a proof that θ is non-vanishing in the upper half-plane.

Corollary 1.4 *If $\text{Im}(\tau) > 0$ and $q = e^{\pi i\tau}$, then*

$$\theta(\tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1})^2.$$

Thus $\theta(\tau) \neq 0$ for $\tau \in \mathbb{H}$.

The next corollary shows that the properties of the function Θ now yield the construction of an elliptic function (which is in fact closely related to the Weierstrass $\wp$ function).

Corollary 1.5 *For each fixed $\tau \in \mathbb{H}$, the quotient*

$$(\log \Theta(z|\tau))'' = \frac{\Theta(z|\tau)\Theta''(z|\tau) - (\Theta'(z|\tau))^2}{\Theta(z|\tau)^2}$$

is an elliptic function of order 2 with periods 1 and τ , and with a double pole at $z = 1/2 + \tau/2$.

In the above, the primes ' denote differentiation with respect to the z variable.

Proof. Let $F(z) = (\log \Theta(z|\tau))' = \Theta(z|\tau)' / \Theta(z|\tau)$. Differentiating the identities (ii) and (iii) of Proposition 1.1 gives $F(z+1) = F(z)$, $F(z+\tau) = F(z) - 2\pi i$, and differentiating again shows that $F'(z)$ is doubly periodic. Since $\Theta(z|\tau)$ vanishes only at $z = 1/2 + \tau/2$ in the fundamental parallelogram, the function $F(z)$ has only a single pole, and thus $F'(z)$ has only a double pole there.

The precise connection between $(\log \Theta(z|\tau))''$ and $\wp_\tau(z)$ is stated in Exercise 1.

For an analogy between Θ and the Weierstrass σ function, see Exercise 5 of the previous chapter.

1.1 Further transformation laws

We now come to the study of the transformation relations in the τ -variable, that is, to the modular character of Θ .

Recall that in the previous chapter, the modular character of the Weierstrass $\wp$ function and Eisenstein series E_k was reflected by the two transformations

$$\tau \mapsto \tau + 1 \quad \text{and} \quad \tau \mapsto -1/\tau,$$

which preserve the upper half-plane. In what follows, we shall denote these two transformations by T_1 and S , respectively.

When looking at the Θ function, however, it will be natural to consider instead the transformations

$$T_2 : \tau \mapsto \tau + 2 \quad \text{and} \quad S : \tau \mapsto -1/\tau,$$

since $\Theta(z|\tau + 2) = \Theta(z|\tau)$, but $\Theta(z|\tau + 1) \neq \Theta(z|\tau)$.

Our first task is to study the transformation of $\Theta(z|\tau)$ under the mapping $\tau \mapsto -1/\tau$.

Theorem 1.6 *If $\tau \in \mathbb{H}$, then*

$$(5) \quad \Theta(z|-1/\tau) = \sqrt{\frac{\tau}{i}} e^{\pi i \tau z^2} \Theta(z\tau|\tau) \quad \text{for all } z \in \mathbb{C}.$$

Here $\sqrt{\tau/i}$ denotes the branch of the square root defined on the upper half-plane, that is positive when $\tau = it$, $t > 0$.

Proof. It suffices to prove this formula for $z = x$ real and $\tau = it$ with $t > 0$, since for each fixed $x \in \mathbb{R}$, the two sides of equation (5) are holomorphic functions in the upper half-plane which then agree on the positive imaginary axis, and hence must be equal everywhere. Also, for a fixed $\tau \in \mathbb{H}$ the two sides define holomorphic functions in z that agree on the real axis, and hence must be equal everywhere.

With x real and $\tau = it$ the formula becomes

$$\sum_{n=-\infty}^{\infty} e^{-\pi n^2/t} e^{2\pi i n x} = t^{1/2} e^{-\pi t x^2} \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t} e^{-2\pi n x t}.$$

Replacing x by a , we find that we must prove

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2} e^{-\pi n^2/t} e^{2\pi i n a}.$$

However, this is precisely equation (3) in Chapter 4, which was derived from the Poisson summation formula.

In particular, by setting $z = 0$ in the theorem, we find the following.

Corollary 1.7 *If $\text{Im}(\tau) > 0$, then $\theta(-1/\tau) = \sqrt{\tau/i} \theta(\tau)$.*

Note that if $\tau = it$, then $\theta(\tau) = \vartheta(t)$, and the above relation is precisely the functional equation for ϑ which appeared in Chapter 4.

The transformation law $\theta(-1/\tau) = (\tau/i)^{1/2} \theta(\tau)$ gives us very precise information about the behavior when $\tau \rightarrow 0$. The next corollary will be used later, when we need to analyze the behavior of $\theta(\tau)$ as $\tau \rightarrow 1$.

Corollary 1.8 *If $\tau \in \mathbb{H}$, then*

$$\begin{aligned} \theta(1 - 1/\tau) &= \sqrt{\frac{\tau}{i}} \sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2 \tau} \\ &= \sqrt{\frac{\tau}{i}} (2e^{\pi i \tau/4} + \cdots). \end{aligned}$$

The second identity means that $\theta(1 - 1/\tau) \sim \sqrt{\tau/i} 2e^{i\pi\tau/4}$ as $\text{Im}(\tau) \rightarrow \infty$.

Proof. First, we note that n and n^2 have the same parity, so

$$\theta(1 + \tau) = \sum_{n=-\infty}^{\infty} (-1)^n e^{i\pi n^2 \tau} = \Theta(1/2|\tau),$$

hence $\theta(1 - 1/\tau) = \Theta(1/2| - 1/\tau)$. Next, we use Theorem 1.6 with $z = 1/2$, and the result is

$$\begin{aligned} \theta(1 - 1/\tau) &= \sqrt{\frac{\tau}{i}} e^{\pi i \tau/4} \Theta(\tau/2|\tau) \\ &= \sqrt{\frac{\tau}{i}} e^{\pi i \tau/4} \sum_{n=-\infty}^{\infty} e^{\pi i n^2 \tau} e^{\pi i n \tau} \\ &= \sqrt{\frac{\tau}{i}} \sum_{n=-\infty}^{\infty} e^{\pi i(n+1/2)^2 \tau}. \end{aligned}$$

The terms corresponding to $n = 0$ and $n = -1$ contribute $2e^{\pi i \tau/4}$, which has absolute value $2e^{-\pi t/4}$ where $\tau = \sigma + it$. Finally, the sum of the other terms $n \neq 0, -1$ is of order

$$O\left(\sum_{k=1}^{\infty} e^{-(k+1/2)^2 \pi t}\right) = O(e^{-9\pi t/4}).$$

Our final corollary of the transformation law pertains to the **Dedekind eta function**, which is defined for $\text{Im}(\tau) > 0$ by

$$\eta(\tau) = e^{\frac{\pi i \tau}{12}} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau}).$$

The functional equation for η given below will be relevant to our discussion of the four-square theorem, and in the theory of partitions.

Proposition 1.9 *If $\text{Im}(\tau) > 0$, then $\eta(-1/\tau) = \sqrt{\tau/i} \eta(\tau)$.*

This identity is deduced by differentiating the relation in Theorem 1.6 and evaluating it at $z_0 = 1/2 + \tau/2$. The details are as follows.

Proof. From the product formula for the theta function, we may write with $q = e^{\pi i \tau}$,

$$\Theta(z|\tau) = (1 + qe^{-2\pi iz}) \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1}e^{2\pi iz})(1 + q^{2n+1}e^{-2\pi iz}),$$

and since the first factor vanishes at $z_0 = 1/2 + \tau/2$, we see that

$$\Theta'(z_0|\tau) = 2\pi i H(\tau), \quad \text{where } H(\tau) = \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau})^3.$$

Next, we observe that with $-1/\tau$ replaced by τ in (5), we obtain

$$\Theta(z|\tau) = \sqrt{i/\tau} e^{-\pi i z^2/\tau} \Theta(-z/\tau | -1/\tau).$$

If we differentiate this expression and then evaluate it at the point $z_0 = 1/2 + \tau/2$, we find

$$2\pi i H(\tau) = \sqrt{i/\tau} e^{-\frac{\pi i}{4\tau}} e^{-\frac{\pi i}{2}} e^{-\frac{\pi i \tau}{4}} \left(\frac{-2\pi i}{\tau} \right) H(-1/\tau).$$

Hence

$$e^{\frac{\pi i \tau}{4}} H(\tau) = \left(\frac{i}{\tau} \right)^{3/2} e^{-\frac{\pi i}{4\tau}} H(-1/\tau).$$

We note that when $\tau = it$, with $t > 0$, the function $\eta(\tau)$ is positive, and thus taking the cube root of the above gives $\eta(\tau) = \sqrt{i/\tau} \eta(-1/\tau)$; therefore this identity holds for all $\tau \in \mathbb{H}$ by analytic continuation.

A connection between the function η and the theory of elliptic functions is given in Problem 5.

2 Generating functions

Given a sequence $\{F_n\}_{n=0}^\infty$, which may arise either combinatorially, recursively, or in terms of some number-theoretic law, an important tool in its study is the passage to its **generating function**, defined by

$$F(x) = \sum_{n=0}^{\infty} F_n x^n.$$

Often times, the defining properties of the sequence $\{F_n\}$ imply interesting algebraic or analytic properties of the function $F(x)$, and exploiting these can eventually lead us back to new insights about the sequence $\{F_n\}$. A very simple-minded example is given by the Fibonacci sequence. (See Exercise 2). Here we want to study less elementary examples of this idea, related to the Θ function.

We shall first discuss very briefly the theory of partitions.

The **partition function** is defined as follows: if n is a positive integer, we let $p(n)$ denote the numbers of ways n can be written as a sum of positive integers. For instance, $p(1) = 1$, and $p(2) = 2$ since $2 = 2 + 0 = 1 + 1$. Also, $p(3) = 3$ since $3 = 3 + 0 = 2 + 1 = 1 + 1 + 1$. We set $p(0) = 1$ and collect some further values of $p(n)$ in the following table.

n	0	1	2	3	4	5	6	7	8	$\dots$	12
$p(n)$	1	1	2	3	5	7	11	15	22	$\dots$	77

The first theorem is Euler's identity for the generating function of the partition sequence $\{p(n)\}$, which is reminiscent of the product formula for the zeta function.

Theorem 2.1 *If $|x| < 1$, then $\sum_{n=0}^{\infty} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$.*

Formally, we can write each fraction as

$$\frac{1}{1-x^k} = \sum_{m=0}^{\infty} x^{km},$$

and multiply these out together to obtain $p(n)$ as the coefficient of x^n . Indeed, when we group together equal integers in a partition of n , this partition can be written as

$$n = m_1 k_1 + \dots + m_r k_r,$$